

Comparative Analysis of Classifiers for EMG Signals

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Abstract—Electromyographic signals have a considerable importance in robotic hand prosthesis and various biomedical applications. The analysis of these signals for pattern recognition of arm movements is helpful to facilitate the handicap individuals with upper limb impairment or paralysed individuals who are able to reinstitute innate control of hand. These signals need to be recorded from the patients hand with the help of electrodes which may be contaminated with noises or undesired signals which affects the output accuracy. To ensure the detection of data with reduced noise and to execute the optimal performance from the analysis, the signals are preprocessed. The data collected for the 52 movements from 27 different subjects is provided by NinaPro database which allows the whole research community to add more advancement to this field. The purpose of this research is to analyse the dataset from this easily accessible database for twelve finger and hand movements acquired from 27 subjects. This processed data was then tested for two different classifiers, Linear Discriminant Analysis classifier and Artificial Neural Network classifier, to examine their percent classification accuracy. The data classified with Linear Discriminant Analysis gives the mean classification accuracy of 85.41% while Artificial Neural Network classifier shows 91.14%. The results for the dataset used in this study revealed that Artificial Neural Network performs better in the classification and recognition of data for hand movements as compared to Linear Discriminant Analysis.

Index Terms—electromyographic signals, classifiers; linear discriminant analysis, artificial neural network, myoelectric control.

I. INTRODUCTION

Measuring the electrical signals generated by muscle activity which includes contractions, relaxation or rest, is known as electromyography (EMG). The electrical signals generated by brain due to the difference in electrical potential between specific cells, are transmitted to the muscles. EMG is able to sense if these signals fail to be sent properly or muscle fails to function properly. EMG uses electrodes to detect the electrical activity of the arm and then different pattern recognition techniques are applied for the classification of movement performed by the subject which help to design a device that is used to help the disabled patients to perform daily life activities like other normal people. Two methods for the prostheses of patients with low-level hand disabilities have been introduced which include electrically powered and myoelectric control. The reliability of EMG signals must be

ensured in such disabilities as hand and wrist movements may be affected by intrinsic muscles and these signals may fail to act as good control signals so optimal function of hand movements must be preserved [1]–[4].

The current research advancement in classification and recognition control of hand prostheses helps the patients to fulfil grasping tasks more easily by using surface EMG signals generated to classify patients goal [5], [6]. The accuracy results for average classification recorded for 10 movements are below 80-90% [7], and may have maximum values up to 90% [8], [9]. At early stages, they started with open close grasp control but now they are able to carry out many functions with hand prostheses. As studies were conducted for limited number of subjects, i.e. 11 healthy and 6 amputated subjects [10], and small number of movements i.e. about 12, so the results obtained might be uncertain. For large number of movements considered, each of them must be recognized with highest accuracy to carry out the real life tasks as desired perfectly. Thus most recent publicly available database named NinaPro (Non Invasive Adaptive Prosthetics) database is based on the data of 52 upper limb movements repeated many times for some time span collected from 27 intact subjects [11], [12]. It consists of EMG information of more number of subjects as compared to other datasets and also uses data obtained from healthy subjects to ensure more accuracy [13].

It is essential to determine which classifier is more accurate in classifying different limb movements. Investigators used several standard approaches for classification purpose, such as Linear Discriminant Analysis (LDA) [14]–[16], Artificial Neural Networks (ANN), k-Nearest Neighbours [17] etc. and recently Support Vector Machine (SVM) [9], [15], [18] was explored. Hudgins et al, for the first time, inspected that data of EMG signals transients that lead to muscle contractions can be utilized for differentiating hand movements by using time domain features. He worked on a non-linear ANN classifier to identify four distinct movements [19]. But other studies conducting comparisons revealed that various classification approaches give different performance results [20]. Although both linear and non-linear classifiers work significantly but LDA classifier is considered to be the frequently used one because of its better classification accuracy as well as more

efficient computationally. The reason for extracting variable results may be due to difference in the experimental setup as efficient non-linear classifiers may have improved performance as compared to linear classifiers with increase in the number of movements to be observed.

This research is focused on the analysis of EMG data by comparing two classification techniques, LDA and ANN, for 12 finger movements to figure out more robust classifier aiming to enhance the efficiency and workability of EMG prostheses system. Methodology of this work is explained in Section II and results are demonstrated in Section III. The general framework of this work is presented in Fig. 1.

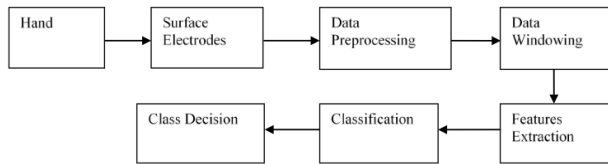


Fig. 1. General framework of the stepwise methodology

II. METHODOLOGY

A. Nina Pro Database

Muscle activity is monitored by using differential electrodes which gives a preprocessed and rectified form of raw sEMG signal. These electrodes use inbuilt frequency filtering techniques which are helpful to overcome high and low frequency transitions. Another combination of 12 electrodes consists of a base station which sample raw sEMG signal at 2kHz.

The Ninapro database cover 10 repetitions of 52 different hand movements (shown in Fig. 2) obtained from 27 intact subjects as shown in figure 2. These movements are divided into four main classes which include 1-12 finger motions, 8 hand positions, 9 Wrist angles and 23 grasping and activity movements. These four classes are subdivided into three exercises (second and third classes are combined into second exercise)

- Finger movements.
- Isometric hand and wrist postures.
- Grasping and activity positions.

This paper is emphasized on the analysis of first exercise of movements. Each movements duration is five seconds with a gap of three seconds of rest. First exercise is completed in 16 minutes, second takes 23 minutes and third lasts for 31 minutes. The dataset for each exercise is comprised of a Matlab file having synchronized variables such as EMG, glove, repetition, stimulus etc. for each subject [11].

During the experiment, subjects sit in front of a laptop with their hand on the desktop. Visual stimuli are received from laptop for each movement and data is recorded simultaneously from the measuring systems. The healthy subjects copy the movements in the video displayed on the laptop screen using their right hand. The disabled patients try to follow the same movements using their damaged hand [21]



Fig. 2. The 52 movements considered in the NinaPro dataset [11]

B. Signal Processing

EMG signals may be contaminated by various noise factors such as causative factors which include extrinsic factors caused by improper electrode separation and electrode position etc. and intrinsic factors i.e. environmental or anatomical etc. The noise factor caused by the environment must be reduced by increasing the signal to noise ratio (SNR) of the EMG signal to get more useful signal [22]. Similarly the proper instalment of electrodes and circuitry may help to overcome the noise induced by electromagnetic devices. If eliminating noise from EMG signals is ignored, desired results cannot be achieved.

After recording, all the data from the database are levelled up to the highest frequency i.e. 100 Hz and then low pass filtered at 1 Hz. In order to remove irrelevant data stored during switching from rest to movement state, data of each movement is segmented into three equal parts and the information from the middle segment is collected. Then the data of the middle segment is further processed by taking its average and one sample for each movement repetition is selected. Thus ten samples are collected for each movement with ten repetitions. The data obtained for each subject is then normalized at zero mean for sEMG signals having unit standard deviation [11].

C. Feature Extraction

Next step in the analysis of sEMG signals is features extraction. This technique is used to obtain appropriate set of features known as feature vector by collecting pertinent information and refusing extraneous data from the raw EMG data. To determine the accurate features representing desired output tasks, feature vector obtained should have all the useful and efficient information derived from the EMG signals. If these signals are subjected to classification methods without processing them carefully with features extraction, their output performance will not be satisfactory and will be inefficient computationally. So this step is of high importance using these features as input to the classifier to ensure better classification

results but their properties such as computational cost, robustness, complexity etc. should never be ignored [23].

Some advanced types of extraction methods are workable with both time and frequency domains but these are computationally inefficient although they contain more useful information [?], [24]. Englehart et al. worked on the combination of time and frequency domain features and found that data obtained from the simultaneous use of the two domains is more useful in recognizing EMG patterns [14], [16].

Time domain features are more feasible to use as they are computationally efficient and less complex compared to frequency domain features [26]. Time-domain features are more helpful to be used for dimensionality reduction purpose thus are well suited for sEMG pattern recognition. This article deals with the four time domain features extracted for EMG data.

Time domain features introduced by Hudgins et al. include mean absolute value (MAV), zero crossing (ZC), slope sign change (SSC) and waveform length feature (WL) [16], [19], [27] known as *Hudgins feature*; are used in this research.

III. CLASSIFICATION ANALYSIS AND RESULTS

The classification of EMG signals is vital for diagnosis of various diseases and needs to be simple, fast and reliable. The high classification accuracy is very crucial as it makes the life of patients having limb disability much more comfortable for which selection of quality features for extraction is of great significance [28]. Furthermore, effective dimensionality reduction techniques and suitable classifiers must be introduced in order to ensure flawless pattern recognition. Recent studies are widely using LDA classifiers as they are simple and easy to be implemented and trained [16], [29].

LDA is a linear classifier which also acts as dimensionality reduction method. It is helpful in both binary and multi-class classification with definite output variable. Each input variable given to this model should have same variance and the data standardized with 0 mean and unit standard deviation.

Researchers have emphasized the use of neural network classifier for EMG analysis as it can deal with both linear and non-linear properties of the data collected for analysis. ANNs resemble the formation of biological neural networks and are non-linear unsupervised learning mechanism capable of modelling and processing EMG data statistically. Del and Park proposed the ANN as the preferable tool to represent the real-time implementation of EMG which can identify the myoelectric signals more precisely [30]. The NinaPro database 1 is comprised of EMG data detected for 12 finger movements acquired from 27 subjects. The data for one of the subjects is demonstrated in Fig. 3.

In this work, the two classifiers LDA and ANN were used for classification of features extracted from the EMG signals. We report subject-wise accuracy, for each classifier, in Table I. The mean accuracies along with standard deviation are shown in Fig. 4. The results of this study show that the performance of Artificial Neural Networks classifier for the classification of twelve finger movements is comparatively better than the

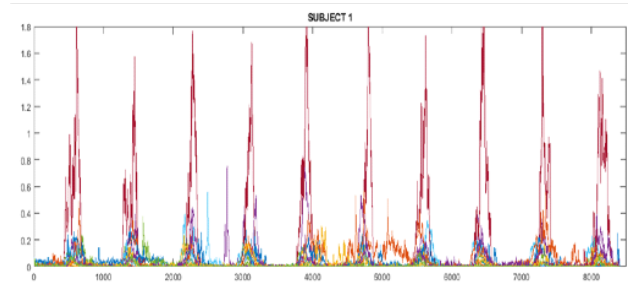


Fig. 3. EMG data obtained for 12 finger movements from Subject 1

TABLE I
CLASSIFICATION ACCURACY FOR EACH SUBJECT

Index	Subjects	ANN Classification (Accuracy %)	LDA Classification (Accuracy %)
1	Subject 1	89.31	81.67
2	Subject 2	94.22	79.01
3	Subject 3	92.10	83.11
4	Subject 4	94.03	85.24
5	Subject 5	95.01	90.32
6	Subject 6	87.43	86.01
7	Subject 7	90.67	81.21
8	Subject 8	93.89	84.76
9	Subject 9	88.47	87.44
10	Subject 10	92.42	91.09
11	Subject 11	91.05	82.72
12	Subject 12	90.11	89.34
13	Subject 13	87.83	87.67
14	Subject 14	94.06	89.89
15	Subject 15	88.35	84.37
16	Subject 16	90.48	83.40
17	Subject 17	93.58	87.05
18	Subject 18	91.69	88.92
19	Subject 19	85.33	81.67
20	Subject 20	92.09	89.98
21	Subject 21	87.49	84.00
22	Subject 22	94.43	86.85
23	Subject 23	91.74	82.34
24	Subject 24	95.01	83.52
25	Subject 25	94.67	86.31
26	Subject 26	86.46	84.76
27	Subject 27	88.92	83.45

results obtained for LDA classifier. The mean classification accuracy attained by ANN classifier is 91.14% and LDA is 85.41%, which is inferior compared to that of ANN classifier.

The statistical significance of above result was established by performing a non-parametric Friedman test. The Friedman test is used to check for differences between groups when the dependent variable being measured is ordinal [31]. The test was performed with null hypothesis that ANN is a better classifier. A non-parametric Friedman test of differences among repeated measures was conducted and rendered a Chi-square value of 18.2 which was significant ($p < 0.05$).

IV. CONCLUSION AND DISCUSSION

The aim of this study is to conclude a precise view of techniques used for processing EMG and to suggest more improvements on the pattern recognition approaches. The

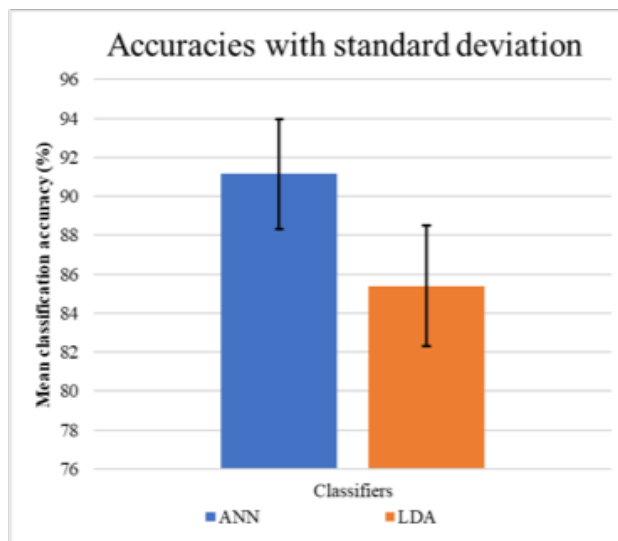


Fig. 4. Mean accuracy for each classifier, computed over 27 subjects. The error bars show standard deviation

comparison will assist researchers to predict the better classifier for the analysis of sEMG signals and the work based on this study will be more helpful in clinical, arm prosthetic and biomedical fields.

This work deals with the analysis of the raw EMG data acquired from the NinaPro database for the identification of twelve finger and hand movements. This publicly available dataset contains data obtained from signals collected with the help of electrodes from the muscle activity of upper limb for ten repetitions of 52 different movements from 27 subjects. Out of 52 movements, twelve finger and hand movements for each subject were selected to be identified for this study. Two classifiers, ANN and LDA were evaluated and compared for the overall percent classification accuracy to identify these movements for four well-known features. The results of this research concluded that non-linear ANN classifier revealed better performance results as compared to linear LDA classifier.

The database used in this study is no doubt very helpful for providing easily available sEMG data, which is applicable efficiently for the recognition and classification of upper limb movements but still there are many challenges faced by bio-robotics community and machine learning research in myoelectric control and recognition. This database maybe upgraded by increasing the number of both healthy and disabled subjects and adding data for dynamic variety of hand movements. This will further boost the functionality and versatility of the database to achieve more-reliable results. In order to meet the challenges confronted in this field, features that are more robust are to be selected and novel features extraction approaches have to be proposed. By increasing the EMG channels count and the number of features as input to the classifier, the count of control commands of the classification technique is also increased. This will help to acquire the remarkable classification outputs by providing more advanced

and significant information from EMG. If large number of features is desired to be extracted, LDA should be preferred because of its dimensionality reduction property as data is converted to a space vector having low dimensions without wasting useful information. Furthermore, the recognition and control of movements maybe enhanced by comparing the classification ability of a variety of linear and non-linear classifiers and the mean accuracy achieved in this study may be further improved.

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